SSU Data Dash 2026: CatBoosters

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Video: https://www.youtube.com/watch?v=96UYkVGed-g

GitHub (code and both submission files): https://github.com/BenIIVVI/datadashstats

Summary

We built two models. The off-market one has no list price to work with, so it values a home in two pieces: what the home is worth at an average market level, and where the market will be in the summer it actually sells. Gradient-boosted trees learn the first piece from 52 features, and our own monthly price index handles the second. At the end we nudge each estimate based on how the model has been doing on nearby homes of the same type. The list-aware model gets to use the list price, and blends a weighted ridge regression with CatBoost, with separate versions for condos and houses.

On walk-forward validation (December 2025 to May 2026, 10,069 homes) the median absolute percentage error is 7.31% off-market and 4.19% list-aware. The organisers' simple baseline gets 17.7%. Most of the error that's left comes from things the data just doesn't record about a home's quality, and that's also why condos are so much easier to price than houses in both models.

1. Data and features

We trained on all 150,000 sales from January 2021 to May 2026 and predicted the 3,974 off-market and 4,064 list-aware test homes, which sold from June to September 2026. Anything you'd only know after a sale (the sale price, sale date, days on market and the external AVM) is taken out of every home we predict. The off-market model never sees the list price at all. The only place we use days on market is from past sales, to guess when a test home is likely to sell.

Market index (off-market)

To compare a sale from 2022 with one from 2025 we needed to take the market out of the price first. We fit a ridge regression of log price on community × type, size, bedrooms, washrooms, parking and a dummy for every sale month, and the month coefficients became our monthly market level. Because it controls for what sold, a month full of condos doesn't look like a crash. The index peaks in March 2022, sits about 25% lower by December 2025, and dips from spring into summer every single year. Every training price gets divided by its month's level, so the trees only have to learn what a home is worth and the index takes care of time.

Off-market features

Each home ends up with 52 features in four groups. The first is the home itself: size (we use the midpoint when it's given as a range, with a flag), bedrooms, washrooms, parking, garage, lot frontage, depth and area, type, style, municipality, community and coordinates. The second is the same building, meaning earlier sales at the exact same coordinates, which we turn into a count and an average adjusted price per square foot. The third is nearby sales, the median adjusted price per square foot of the 10 closest earlier sales of the same type multiplied by this home's size, plus how far away they are. The last is the listing text. A ridge regression on TF-IDF word and word-pair counts predicts how far a home's price per square foot sits above its community and type average, and we add 30 SVD topics on top.

We were careful about leakage here. A training sale only ever sees sales from earlier months, the building numbers leave out the sale itself, and the text scores for training rows are all out-of-fold.

List-aware features

The ridge part of the list-aware model uses the simple numeric features (list price, location, size, rooms, months since January 2021 and property type), standardized with scikit-learn's StandardScaler. CatBoost can take categories as they are, so we also gave it municipality, community, house style, garage type, lot dimensions and a combined community and property type column, which picks up pricing patterns that are specific to one kind of home in one area. We tried adding seasonal features to CatBoost too, but MdAPE got noticeably worse so we took them out.

2. Models

Off-market

The target is the market-adjusted log price, and older sales count less, with a sale's weight halving every 12 months. The value estimate is an average of four histogram gradient-boosting models from scikit-learn. Two use 2,000 trees at a learning rate of 0.025 and the other two use 800 trees at 0.05. Every leaf needs at least 20 sales, each split looks at 30% of the

features, and each model gets its own seed. To give a sense of why we average so many trees, on the 2024 summer one tree on its own missed by 11.1% while the four-model average missed by 6.85%.

The market level for a test home is the projected index averaged over the dates it's likely to sell. We take its listing date, add 100 typical times on market from past sales, and keep the ones that fall between June and September. For months after May we take the last known level plus the median change for that calendar month in earlier years, so one strange year can't drag the whole projection around.

The last step is a local fix. We get an honest out-of-fold miss for every training sale using five folds, then for a new home we average the misses of its 5, 10, 20 and 40 nearest earlier sales of the same type. A ridge regression per property type decides how much to trust each of those four averages. The final price is exp(value + market level + local fix), which means the three pieces multiply in dollar terms. One detached house in the 2024 backtest works out to $898,771 × 0.990 for the June market × 1.121 because its area keeps selling above our guesses, which gives $997,309. It sold for $994,600.

List-aware

Here ridge regression predicts the sale price directly, and CatBoost predicts the log of the sale price divided by the list price, which we convert back to dollars. Condos (apartments, condo townhouses and co-ops) and houses each get their own pair of models. For condos the two are weighted evenly, and for houses it's 30% ridge and 70% CatBoost. Ridge is good at the broad relationships and CatBoost catches the messier patterns. Ridge weights each sale by 1 / price³ and CatBoost by (median price / price)^1.5, which gives cheaper homes more say in training so errors count more evenly in percentage terms.

Why they're different

The list price already tells you a lot. It reflects the current market and whatever the seller's agent knows about the home's condition, so the list-aware model only has to learn how far the sale lands from the list, and doesn't need an index or comparable sales. The off-market model isn't allowed to look at the list price, so it has to rebuild all of that from scratch, and the index, nearby sales, building history, listing text and local fix are basically standing in for it. That's the main reason its error ends up roughly twice as big.

3. Validation

We never used a random split, since that would let the model learn from sales that happened after the ones it's predicting. Instead we tested in two ways.

The numbers we report come from walk-forward validation, which we ran for both tracks. Each month from December 2025 to May 2026 is predicted by a model trained only on sales from before that month started, so that's six full refits per track. For the list-aware model the sale price and sale date are never used as inputs, past prices are only used for training, and the validation prices are only used to score the predictions.

We also ran summer backtests that copy the real test, training on sales before June 1 and predicting June through September. All of our off-market modelling decisions were made on the 2023 and 2024 summers, and we kept 2025 aside. To be upfront, we did look at 2025 once and then again after the last two changes, so we'd call it a confirmation more than a perfectly clean holdout. We also reran 2021 and 2022 off-market with the final model.

4. Results

Walk-forward, Dec 2025–May 2026	Homes	MdAPE % (95% CI)	MAPE % (95% CI)	Median signed error %
Off-market	10,069	7.31 (7.14–7.50)	9.44 (9.28–9.61)	+1.38
List-aware	10,069	4.19 (4.09–4.31)	5.33 (5.24–5.42)	−0.62

Table 1: Walk-forward validation. Brackets show a bootstrap 95% interval, treating homes as independent.

Summer (train before June 1)	2021	2022	2023	2024	2025
Off-market MdAPE / MAPE %	7.56 / 10.07	8.30 / 11.60	6.62 / 8.88	6.64 / 8.96	6.84 / 9.08
Off-market median signed error %	+0.4	+5.1	+0.9	+1.5	+0.9
List-aware MdAPE / MAPE %			5.50 / 6.66	5.06 / 6.21	4.30 / 5.37
List-aware median signed error %			+1.2	+1.4	+0.8

Table 2: Summer backtests. 2021 only had five months of training data, and 2022 was the year prices fell fast.

Off-market, our typical miss of 7.31% compares with 17.7% for the simple baseline and 20.8% for the starter model (the 10 nearest sales). Validation looks a bit worse than the summers because March and April 2026 were tough months at 7.9% and 7.7%, with prices climbing quickly, while the other four months were between 6.8% and 7.2%. Since the hidden test is a summer, we expect something around 6.7 to 7.0% off-market, unless the market turns sharply the way it did in 2022.

List-aware, MdAPE goes from 7.26% for the list-price baseline down to 4.19%, which is about a 42% cut, and it holds steady across the six months at 4.0 to 4.4%. Its summers are worse than its validation though (4.3 to 5.5%, and it tends to guess about 1% high), so for the hidden test we'd honestly expect something closer to 4.5 to 5%. Looking at every off-market prediction we made from earlier sales only, 54,991 of them, 64% land within 10% of the sale price and 89% within 20% (figure 1).

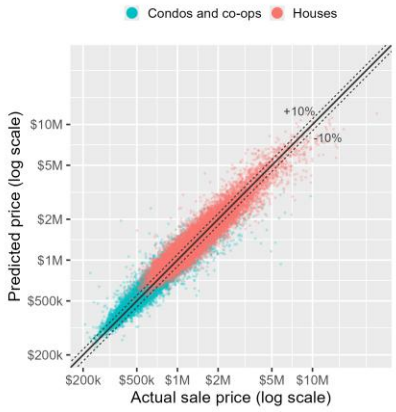


Figure 1: Off-market predicted vs actual price for every out-of-sample prediction, on log scales. The solid line is a perfect guess and the dotted lines are ±10%.

5. Where the model fails

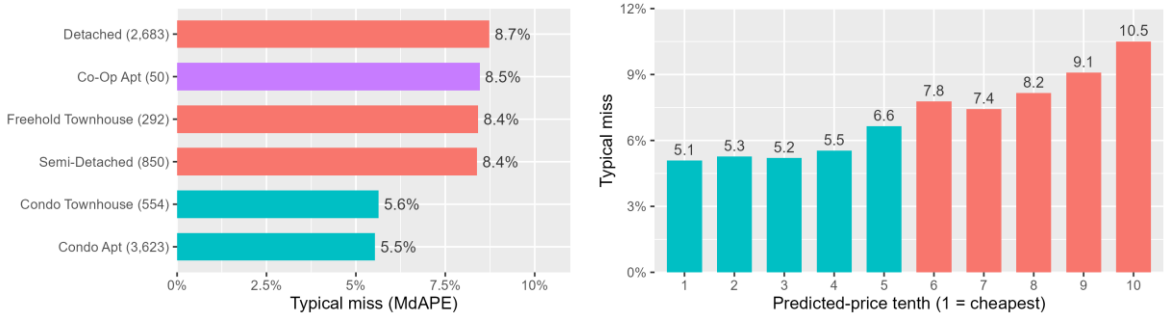


Figure 2: Off-market typical miss on the 2025 summer by property type (number of homes in brackets) and by tenth of predicted price.

Off-market

The biggest split is by property type. Condos miss by about 5.5% and houses by 8.4 to 8.7%. Units in the same condo building are close to copies of each other, so building history pins them down well. Two houses with the same size, rooms and street can still be completely different in age, renovations, lot and finishes, and none of that is in the data. Price works the same way, with the cheapest tenth of homes missing by 5.1% and the most expensive tenth by 10.5%, since expensive homes are rarer, more varied and mostly houses.

Neighbourhoods vary a lot too. Across the 97 communities with at least 40 sales in 2025, the typical miss runs from 3.9% in Cedarglen, which is mostly condos, up to 13.5% in Lindencliffe, where 73% of sales are houses. The middle half of communities fall between 6.1% and 8.0%, and the error goes up where more of the sales are houses (correlation 0.39) and where prices are higher (0.43). There's also a timing effect. Our estimates run about 2% high in July and August when the summer dip goes deeper than we projected, and in 2022, when prices fell after the spring peak, they ran 5% high.

We also wanted to be honest about uncertainty, so each estimate gets an 80% range taken from the 10th to 90th percentile of the 2023 and 2024 misses for its property type. On the 2025 homes that range covered 79.5% of sales, and it's usually about 22.6% of the price wide, or roughly ±11%.

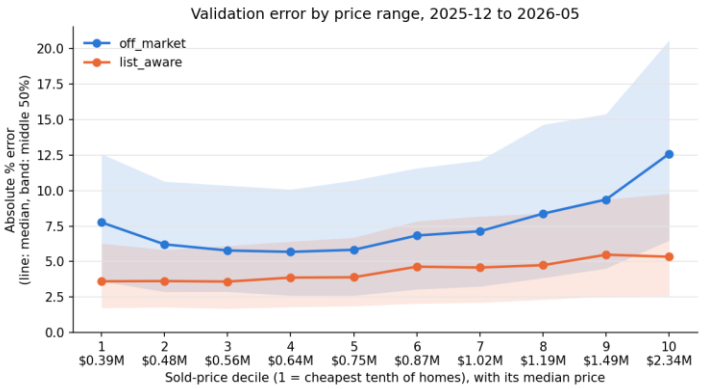


Figure 3: Walk-forward validation error by sold-price decile for both tracks. The line is the median and the band is the middle 50%.

List-aware

Condo apartments are the easiest for the list-aware model at about 3.6%, and semi-detached homes are the hardest at 5.7%. Condo apartments tend to share layouts and building features, while houses differ more in condition, renovations and location. Errors also climb for more expensive homes (figure 3), maybe because those homes have more unusual features, and maybe because training gives cheaper homes more weight, though we haven't tested either explanation. One thing to keep in mind is that ridge tends to guess too low and CatBoost too high, and the blend works partly because the two cancel out.

6. Limitations

Plenty of things we tried for the off-market model didn't help, all judged on the 2023 and 2024 summers. Separate price indexes for houses and condos made things worse, because the extra noise in the projection outweighed the bias it fixed. Following each community's recent price trend didn't work either, since those trends have an autocorrelation of −0.41 and tend to reverse instead of continuing. Blending in a random forest (9.2% on its own) and fitting separate tree models for houses and condos didn't add anything, and neither did a fixed summer correction, because the best amount changed from 0 to 1.75% depending on the year. On the list-aware side, seasonal features in CatBoost made it worse.

We wouldn't trust the off-market model much for expensive or unusual houses, for thin groups like co-ops (only 50 test homes in 2025) or rarely traded communities, or in a summer where the market turns sharply. The list-aware model leans on the list price, which can reflect marketing or a selling strategy rather than real value, and it knows nothing about renovations, staging or the foundation. Neither model can see a sudden market change coming. Data on age, renovations, floor level and condo fees would probably help both more than anything else. If we kept going, we'd test condition data and compare different sample weights and ridge/CatBoost blend weights on the same validation months, especially for expensive homes.

Outside data

None. We didn't use any outside data or sources.

Reproducing our results

From a fresh copy of the repository, run pip install -r requirements.txt (catboost is included, or install it with pip install catboost), then python run.py backtest, validate, report, predict and check. The seeds are fixed, so this rebuilds both submission files exactly.